

Problem-1

We will get max Power for case-IV

TTL (Transistor-Transistor logic)

2 VP TTL NAND gate.

Case-IV

$V_2 = 5V, V_3 = 5V$

EB, E_2B junction $\rightarrow RB$

CB junction $\rightarrow FB$

$T_1 \rightarrow$ Reverse Active

$T_2, T_0 \rightarrow$ Sat. $\rightarrow V_{BE} = 0.8V$

$V_{CE} = 0.1V$

$V_0 = 0.1V$ $V_{B0} = 0.8V$

$V_p = 0.8 + 0.8 = 1.6V$

$V_g = 0.8 + 0.1 = 0.9V$

$V_{B1} = 0.7 + 0.8 + 0.8 = 2.3$

i) Find V_0 , currents and verify that the above ckt implements a NAND Gate.

ii) Find P_T, P_{avg} .

given

$V_{BE(sat)} = 0.8V$ $V_{IL} = 0.1V$ $\beta = 0.1$

$V_{CE(sat)} = 0.1V$ $V_{IH} = 5V$

$I_1 = \frac{5 - 2.3}{4K} = 0.67mA$

$I_2 = \frac{5 - 0.9}{1.6K} = 2.56mA$

$I_3 = \frac{5 - 0.1}{4K} = 1.225$

$\beta = \frac{I_E}{I_B} = \frac{I_{E2}}{I_1}$

$\therefore I_{E2} = \beta \times I_1 = 0.1 \times 0.67 = 0.067mA$

$I_{E1} = \beta \times I_2 = 0.067mA$

$I_{E1} = I_1 + I_{E2} + I_{E3} = 0.804mA$

$I_{E2} = I_{E1} + I_2 = 3.364mA$

$I_k = \frac{0.8 - 0}{1} = 0.8mA$

$I_{B0} = I_{E2} - I_k = 2.564$

$P_T = (5 - 0) \times (I_{E1} + I_2 + I_3)$

$= 22.92mW$

Problem-2

(Set-A)

(a) $K_n = 0.2 \text{ mA/V}^2$, $V_{tn} = 0.6 \text{ V}$

$V_A = V_B = V_{GS}$

$V_{DS} = V_Y$

* Case-I $\rightarrow V_A = 5 \text{ V}$, $V_B = 5 \text{ V}$

$M_A, M_B \rightarrow$ triode region

$I_D = I_{DA} + I_{DB}$

$\Rightarrow \frac{5 - V_Y}{20 \text{ K}} = K_n \left\{ 2(V_{GS} - V_{tn})V_{DS} - V_{DS}^2 \right\} \times 2$

$\Rightarrow (5 - V_Y) = 20 \times 0.2 \times \left\{ 2(5 - 0.6)V_Y - V_Y^2 \right\} \times 2$

$\Rightarrow 5 - V_Y = 8(8.8V_Y - V_Y^2)$

$\Rightarrow 5 - V_Y = 70.4V_Y - 8V_Y^2$

$\Rightarrow 8V_Y^2 - 71.4V_Y + 5 = 0$

Solving $\rightarrow V_Y = 8.854, 0.07 \text{ V}$

$\therefore V_Y = 0.07 \text{ V}$

$I_D = \frac{5 - 0.07}{20 \text{ K}} = 0.2465 \text{ mA}$

$I_{DA} = I_{DB} = \frac{I_D}{2} = 0.123 \text{ mA}$

* Case-II: $V_A = 0 \text{ V}$, $V_B = 0 \text{ V}$
 $M_A, M_B \rightarrow$ Cutoff

$I_D = I_{DA} = I_{DB} = 0 \text{ mA}$

$I_D = 0$

$\frac{V_{DD} - V_Y}{20} = 0 \therefore V_Y = V_{DD}$

$V_Y = 5 \text{ V}$

(b) Power Consumption

* Case-I

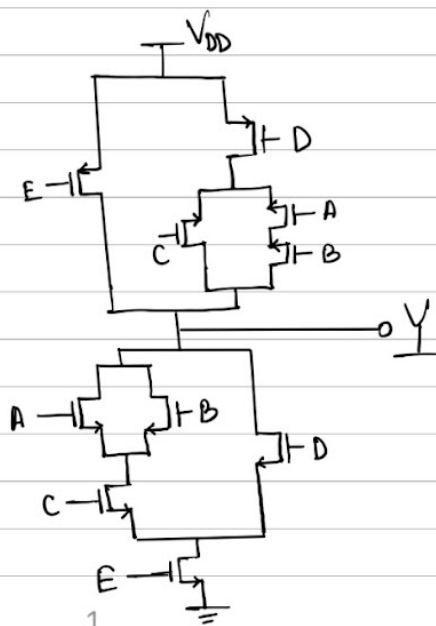
$P_1 = (V_{DD} - 0) I_D$
 $= (5 - 0) \times 0.2465$
 $= 1.23 \text{ mW}$

* Case-II

$P_2 = (V_{DD} - 0) I_D$
 $= 0 \text{ mW}$

(c)

$Y = ((A+B)C+D)E$



(Set-B)

(a) $K_n = 0.2 \text{ mA/V}^2$, $V_{tn} = 0.6 \text{ V}$

$V_A = V_B = V_{GS}$

$V_{DS} = V_Y$

* Case-I $\rightarrow V_A = 6 \text{ V}$, $V_B = 6 \text{ V}$

$M_A, M_B \rightarrow$ triode region

$I_D = I_{DA} + I_{DB}$

$\Rightarrow \frac{6 - V_Y}{15 \text{ k}} = K_n \{ 2(V_{GS} - V_{tn})V_{DS} - V_{DS}^2 \} \times 2$

$\Rightarrow (6 - V_Y) = 15 \times 0.2 \times \{ 2(6 - 0.6)V_Y - V_Y^2 \} \times 2$

$\Rightarrow 6 - V_Y = 6 (10.8 V_Y - V_Y^2)$

$\Rightarrow 6 - V_Y = 64.8 V_Y - 6 V_Y^2$

$\Rightarrow 6 V_Y^2 - 68.8 V_Y + 6 = 0$

Solving $\rightarrow V_Y = 10.87, 0.091 \text{ V}$

$\therefore V_Y = 0.091 \text{ V}$

$I_D = \frac{6 - 0.091}{15} = 0.393 \text{ mA}$

$I_{DA} = I_{DB} = \frac{I_D}{2} = 0.196 \text{ mA}$

* Case-II: $V_A = 0 \text{ V}$, $V_B = 0 \text{ V}$

$M_A, M_B \rightarrow$ Cutoff

$I_D = I_{DA} = I_{DB} = 0 \text{ mA}$

$I_D = 0$

$\frac{V_{DD} - V_Y}{15} = 0 \therefore V_Y = V_{DD}$

$V_Y = 6 \text{ V}$

(b) Power Consumption

* Case-I

$P_1 = (V_{DD} - 0) I_D$
 $= (6 - 0) \times 0.393$
 $= 2.358 \text{ mW}$

* Case-II

$P_2 = (V_{DD} - 0) I_D$
 $= 0 \text{ mW}$

(c) $Y = (AB + C)D + E$

